Weekly Progress Report

Amity School of Engineering & Technology

B. Tech: Program

WPR No: 6	**Duration: 24 August 2026 – 30 August 2026**

Weekly Progress Report (WPR) for Minor Project

To be filled by Students	
Students Name	Rohit Yadav, Vishal, Hardik Mehlawat (Group 148)
Roll no.	472 (Rohit), 568 (Vishal), 449 (Hardik)
Enrollment no.	A2305223472 (Rohit), A2305223568 (Vishal), A2305223449 (Hardik)
Project Title finalized, if Yes, give name, if NO, give reason	Yes — Deep Learning for AI-Generated Image Detection: A Review of CNN, Vision Transformer, and Explainable Approaches
Synopsis submitted	Submitted (on Amizone, with offer letter)
Literature review	Extraction complete for all 48 primary studies; taxonomy of detection approaches finalised; all 74 reference identifiers verified against arXiv/Crossref records and BibTeX file built
Technical & Economical Feasibility	Not started
Bill of Material	N/A (literature review project this semester)
Project Progress Schedule (PERT Chart)	Week-wise schedule prepared; PERT chart pending
Design of critical components	Paper structure fixed (8 sections); Section 3 taxonomy finalised across subsections 3.1–3.5
Fabrication work (give %)	N/A
Experimental work (give %)	Not started (planned for Semester 8)
Result and Analysis	Not started
Report writing	In progress — Sections 1, 2 and 3.1–3.2 drafted (about 3,470 words); Sections 3.3–3.5 next
Signature of students	

Work done in this week

Sections 3.1 and 3.2 of the review paper were drafted in full this week, adding about 1,504 words to the Section 3 file. A lead paragraph of 159 words was written first, introducing the five-family taxonomy used across Sections 3.1 to 3.5 and stating the order in which the families are presented. Section 3.1, on handcrafted, statistical and frequency-domain forensics, runs to 645 words and covers all 14 primary studies assigned to it: the spectral cues left by repeated upsampling, how fragile those cues become once an image is compressed or the generator's architecture is redesigned, learned frequency front-ends that replace hand-chosen filters, and fingerprint and co-occurrence based approaches. Section 3.2, on CNN-based approaches and transfer learning, runs to 700 words and covers all 19 studies assigned to it: the ProGAN-trained transfer-learning baseline, patch-level classification, universal GAN detection, the accuracy collapse these detectors show when tested on diffusion-generated images, methods that classify a derived input such as a gradient map or a reconstruction error instead of the image itself, and innovations in the training signal. Forty-one verified references are now cited in Section 3, and every accuracy and dataset figure reported in the two subsections was copied from the extraction table rather than restated from memory. One extraction-table cell was clarified during the internal review: the Frank et al. study analyses six GANs overall but uses four of them in its classification experiments, and the wording was corrected to match. With Sections 1 (939 words), 2 (1,024 words) and 3.1–3.2 (1,504 words) complete, the draft now stands at roughly 3,470 words. Sections 3.3 to 3.5 — transformer and attention-based, multimodal and foundation-model-based, and explainability approaches — are scheduled next.

To be filled by Guide (strike off whichever is not applicable)	
Performance of students is satisfactory [ ]	
Performance of students is unsatisfactory [ ]	
A warning to be issued to student(s) (Name) ______________________________	
Student was not well (Name) ______________________________	
Date: 2 September 2026	**Signature of Guide**

Date: 2 September 2026